

Bank Customer Churn Behavior Analysis Report

An end-to-end data analytics project using Python, SQL, and Streamlit.

Metric	Value
Total Customers	10,000
Churned Customers	2,038
Overall Churn Rate	20.38%
Avg Balance (Churned)	\$91,109.48
Avg Balance (Retained)	\$72,742.75
Avg Age of Churned Customer	44.8
High-Value Churned	592

1. Project Overview

This project performs an industry-standard end-to-end data analysis of bank customer churn behavior using a dataset of 10,000 customers across data cleaning, feature engineering, SQL analysis, and interactive dashboard reporting.

Phase	Tool	Purpose
Data Preparation	Python (Pandas, NumPy)	Cleaning, EDA, Feature Engineering
Data Analysis	SQL (SQLite)	Business Question Answering
Visualization	Plotly + Streamlit	Interactive Dashboard
Reporting	ReportLab	PDF Report Generation

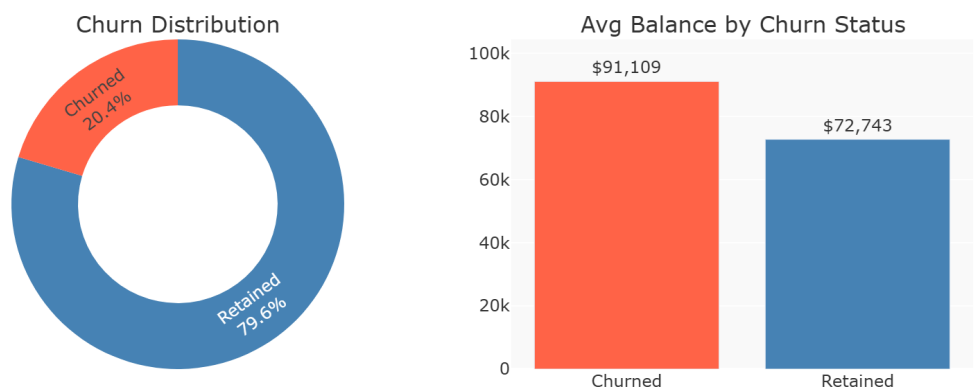
2. Dataset Summary

The dataset contains 10,000 bank customers with 18 features covering demographics, financial information, and behavioral attributes. No missing values were present after preprocessing. Key engineered features include age_group, tenure_bucket, value_score, and balance_salary_ratio.

3. Key Findings

3.1 Overall Churn Distribution

The overall churn rate is 20.38%, with 2,038 customers out of 10,000 having exited. Churned customers hold a higher average balance (\$91,109.48) than retained ones (\$72,742.75), suggesting financial dissatisfaction rather than inability to maintain accounts.

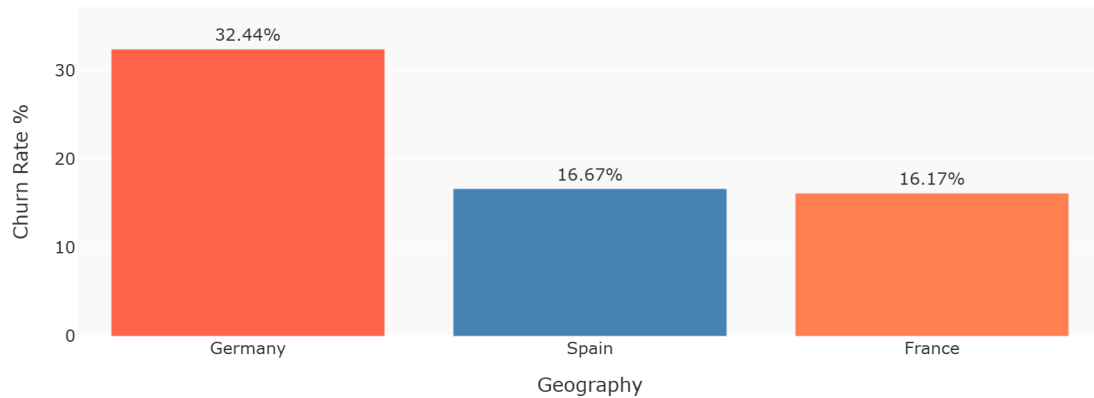


3.2 Churn by Geography

Germany shows a significantly higher churn rate compared to France and Spain, suggesting region-specific service or competitive issues.

Geography	Total Customers	Churned	Churn Rate
Germany	2509	814	32.44%
Spain	2477	413	16.67%
France	5014	811	16.17%

Churn Rate by Geography

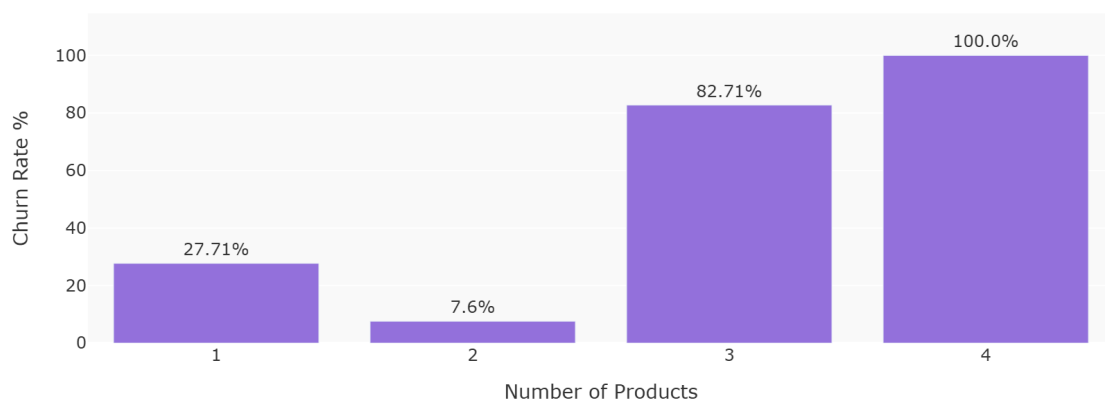


3.3 Churn by Number of Products

Customers with 3-4 products churn at an extremely high rate, suggesting over-selling leads to dissatisfaction.

No. of Products	Total	Churned	Churn Rate
1.0	5084.0	1409.0	27.71%
2.0	4590.0	349.0	7.6%
3.0	266.0	220.0	82.71%
4.0	60.0	60.0	100.0%

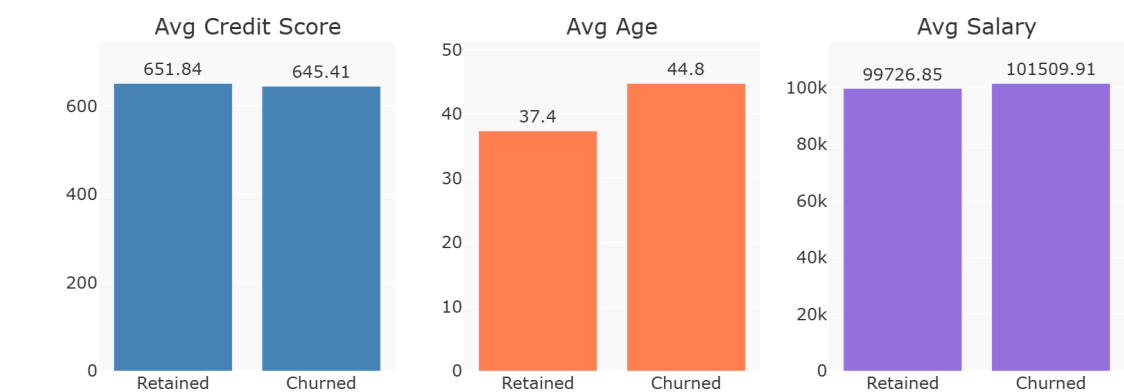
Churn Rate by No. of Products



3.4 Avg Metrics: Churned vs Retained

Churned customers are on average older, with similar salary levels but notably different balance profiles compared to retained customers.

Status	Avg Balance	Avg Salary	Avg Credit Score	Avg Age
Retained	\$72,742.75	\$99,726.85	651.84	37.4
Churned	\$91,109.48	\$101,509.91	645.41	44.8

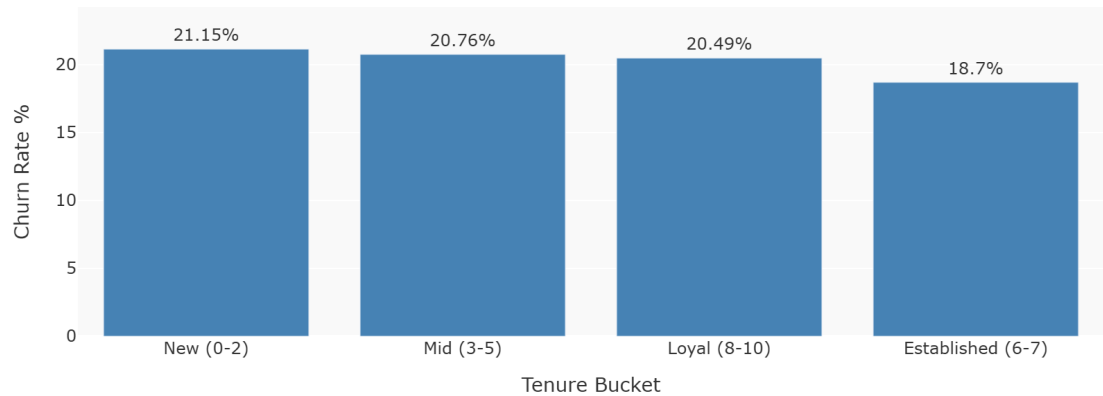


3.5 Churn by Tenure Bucket

Churn is relatively consistent across tenure groups, indicating that even long-term customers are not immune to attrition.

Tenure Bucket	Total	Churned	Churn Rate
New (0-2)	2496	528	21.15%
Mid (3-5)	3010	625	20.76%
Loyal (8-10)	2499	512	20.49%
Established (6-7)	1995	373	18.7%

Churn Rate by Tenure Bucket

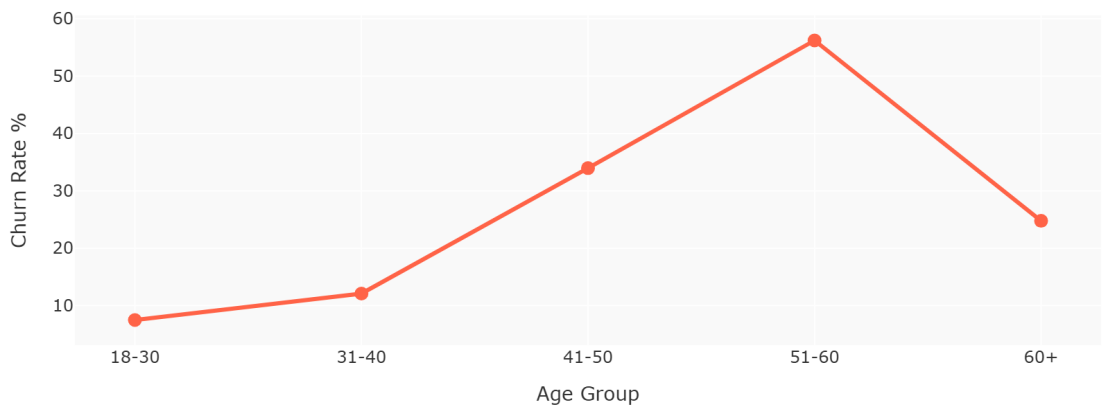


3.6 Churn by Age Group

The 41-50 age group shows the highest churn rate, indicating mid-career customers are most likely to switch banks.

Age Group	Total	Churned	Churn Rate
nan	22	2	9.09%
18-30	1946	146	7.5%
31-40	4451	539	12.11%
41-50	2320	788	33.97%
51-60	797	448	56.21%
60+	464	115	24.78%

Churn Rate by Age Group

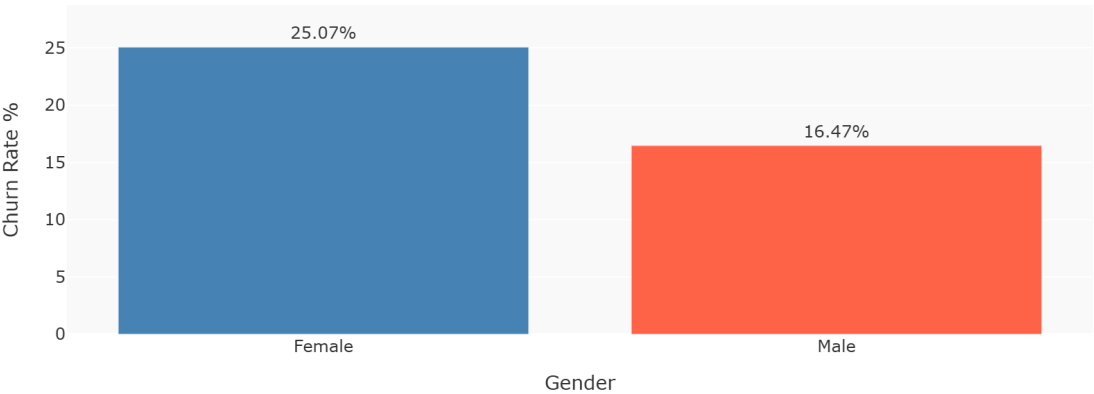


3.7 Churn by Gender

Female customers churn at a higher rate than male customers, suggesting gender-specific needs may not be fully addressed.

Gender	Total	Churned	Churn Rate
Female	4543	1139	25.07%
Male	5457	899	16.47%

Churn Rate by Gender



3.8 Top 10 High-Value At-Risk Customers

592 high-value customers have already churned. The table and chart below show the top 10 currently retained customers most at risk based on their composite value score.

Customer ID	Surname	Geography	Age	Balance	Value Score
15571958	McIntosh	Spain	40	\$221,532.80	0.8735
15780212	Mao	France	37	\$212,692.97	0.8614
15677764	Chao	Germany	74	\$186,445.31	0.8394
15747174	Hao	Germany	58	\$190,298.89	0.8376
15651103	Sal	Spain	69	\$183,744.98	0.8334
15799534	McClaran	France	71	\$183,135.39	0.8333
15604782	Tan	Germany	33	\$187,257.94	0.8287
15712409	Tang	Germany	66	\$182,532.23	0.8274
15762331	Moss	France	37	\$178,755.84	0.8274
15573319	Azubuike	Germany	35	\$178,317.60	0.8213

Top 10 High-Value At-Risk Customers



4. Business Recommendations

1. Target Germany with retention campaigns

Germany has the highest churn rate. Introduce loyalty rewards, personalized offers, or dedicated relationship managers for German customers.

2. Re-engage inactive members

Inactive members churn at a significantly higher rate. Trigger re-engagement campaigns via email or app notifications with personalized incentives.

3. Audit multi-product customers

Customers with 3-4 products show extremely high churn. Review the product bundling strategy and ensure customers are not being over-sold.

4. Prioritize high-value at-risk retention

592 high-value customers have already churned. Implement an early warning system using `value_score` to flag at-risk customers.

5. Focus on the 41-50 age segment

Middle-aged customers show the highest churn rates. Tailor premium offerings to meet their higher expectations.

5. Conclusion

This project demonstrates a complete, industry-standard data analytics workflow applied to customer churn in banking. Through Python EDA, SQL analysis, and an interactive Streamlit dashboard, we identified key churn drivers including geography, product count, age group, and activity level. The engineered features provide a strong foundation for future machine learning churn prediction models.